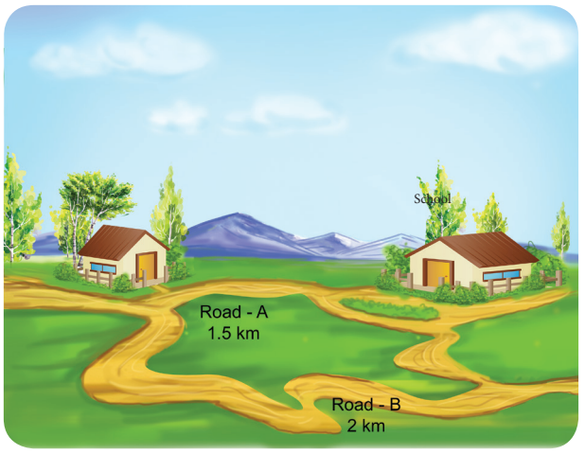
Introduction

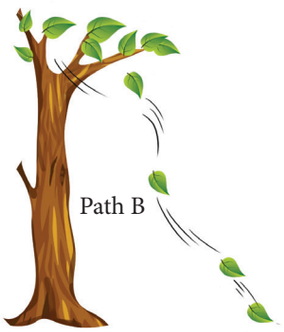
Look at the picture given below. Kavitha can reach her school in two ways, as shown in the picture. Can you tell, by choosing which path she could reach the school early?



In the picture given below, you can see leaf falling from a tree. In which path the leaf will reach the ground first?



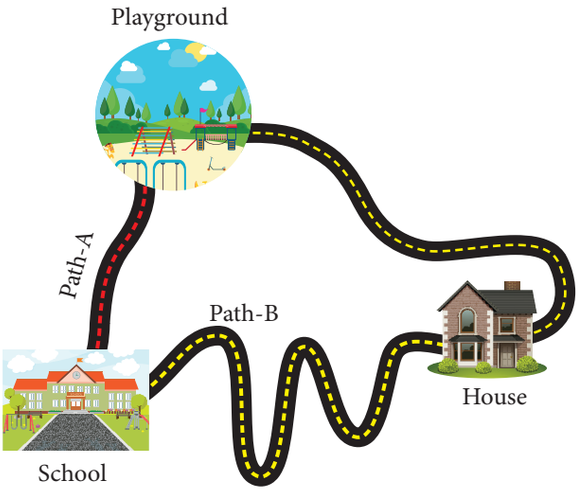
Path of the leaf, when there is no wind.



Path of the leaf when there is wind.

Uma and Priya are friends studying in the same school. After school hours, they go to the nearby playground, play games and return back home. One day Uma told that she would reach the playground after visiting her grandmother's house. The paths which they took to reach the playground is shown here.

Take a twine and measure the length of the two paths (A and B). Which is the longest path among the two?



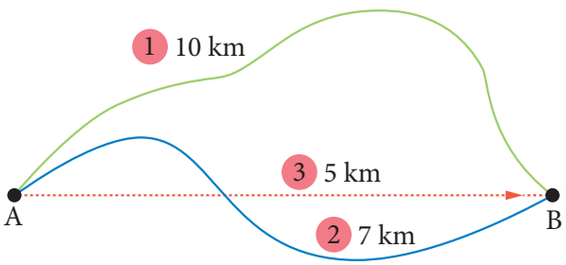
From the above examples, we could conclude that when an object travels from one place to another, it will reach faster if it travels along the straight line path. The straight line path is the shortest distance between two points.

In this lesson we are going to study about distance and displacement, speed and velocity, acceleration, distance - time graph, velocity - time graph, centre of gravity and stability.

2.1 Distance and Displacement

The total length of a path taken by an object to reach one place from another place is called distance. The shortest distance from the initial position to the final position of an object is called displacement. Both distance and displacement possess the same unit. The SI unit distance and displacement is metre (m).

The figure given below shows the motion of a person between two places A and B.



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